

# A Real-Time Traffic Digital Twin from Public CCTV: Vision-Based Detection, Map-Anchored Localization, and Self-Calibrating Speed Estimation

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**Abstract**—We present a real-time traffic digital twin that converts live public CCTV feeds into an interactive, map-anchored view of road traffic. The system detects vehicles with YOLOv26, tracks them with BoxMOT, and projects each pixel position to a geographic coordinate via a perspective homography. Because public CCTV cameras are uncalibrated, accurate ground-plane mapping and speed estimation are the central challenges. We address them with three mechanisms: (i) a two-stage curved transform that maps pixels along the true road centreline from the national node-link road network; (ii) a five-stage speed estimator with physics-based outlier rejection, sliding-window least-squares regression, and per-track exponential smoothing; and (iii) geometric recovery of the absolute speed scale via a *focal-free* pose solve that reads the focal length from the dashed lane-marking period. The homography is produced without manual setup by a warm-up pipeline that builds a vehicle-free *clean plate* to expose those markings and solve camera pose. An ITS-referenced segment-speed cross-check is available as an optional refinement but is disabled by default. The system additionally supports multi-camera background monitoring, congestion clustering, and a SQLite time-series history. We describe the architecture and key algorithms, and report latency, throughput, tracking stability, and localization accuracy over live streams.

**Index Terms**—digital twin, intelligent transportation systems, object detection, multi-object tracking, homography, perspective transform, speed estimation, real-time visualization.

## I. INTRODUCTION

Cities expose thousands of public traffic cameras, but the raw feeds are difficult to interpret at scale: an operator watching a video grid cannot read off vehicle counts, speeds, or where congestion is forming on a map. A *digital twin*—a live, geo-referenced model of the road state—turns those feeds into actionable information. The technical obstacle is that public CCTV cameras are *uncalibrated*: their height, pitch, focal length, and exact heading are unknown, so projecting an image detection onto the ground plane (and therefore measuring distance and speed) is ill-posed without per-camera calibration.

This paper describes a system that ingests Korean ITS CCTV streams and produces a real-time map-anchored twin. Our contributions, from a systems and algorithms standpoint, are:

- **Map-anchored localization.** We snap each camera onto the national node-link road network, extract the road centreline, and map pixels along the *curve* of the road

in two stages, rather than assuming a flat homographic plane. Satellite-referenced landmarks (Section V) show the pipeline places vehicles with 1.8–10.3 m lateral (on-road) accuracy; the curved stage constrains markers to the mapped road shape on bends without degrading an already-accurate flat fit.

- **Robust speed estimation without ground truth.** A five-stage filter (duplicate skip, physics jump guard, sliding-window least-squares regression, per-track exponential smoothing with spike rejection, and scale correction) produces stable speeds despite detection jitter and tracker ID churn.
- **Geometric speed-scale recovery.** The absolute longitudinal (speed) scale is recovered *geometrically* by a focal-free pose solve that reads the focal length from the dashed lane-marking period on the warm-up clean plate, removing the dependence on any external speed reference. An ITS-referenced `speed_scale` cross-check remains available as an *optional, off-by-default* refinement for cameras where dashed markings are not observable.
- **A complete, resilient pipeline.** A warm-up clean-plate auto-calibration that solves camera pose with focal-free longitudinal anchoring and locks once committed (manual 4-point calibration available), dual-path inference (browser canvas and server-side HLS) over a shared detector, multi-camera background monitoring, congestion clustering, and a historical analytics store.

The rest of the paper is organized as follows. Section II reviews related work and positions this system relative to the state of the art. Section III presents the system architecture. Section IV details the algorithms. Section V reports the measurements. Section VI discusses limitations, and Section VII concludes.

## II. RELATED WORK

**Object detection and tracking.** Modern traffic vision builds on real-time detectors. NMS-based architectures such as YOLOv8 [1] suppress redundant boxes through non-maximum suppression, but the threshold-sensitivity of NMS causes bounding-box positions to jitter between frames; this per-frame

noise propagates directly into downstream speed estimates. To eliminate the source of positional instability, we adopt YOLO26 [2], an NMS-free end-to-end detector that produces more consistent bounding-box positions and simplifies the TensorRT FP16 engine graph by removing the NMS layer [3], lowering both inference latency and its variance; YOLOv8 is retained as a legacy fallback. For tracking we use the BoxMOT framework [4], which exposes ByteTrack [5], BoT-SORT [6], and OC-SORT [7]; appearance re-identification uses OSNet [8]. Trackers without appearance ReID suffer ID switches under occlusion, which we mitigate with a lightweight position-based ID stabilizer.

**Camera-to-ground mapping.** Mapping image points to a ground plane is a homography problem [9], typically solved from four correspondences with OpenCV [10]. For *uncalibrated* cameras, vanishing-point and lane-geometry cues are commonly used to recover pose [11], but these techniques assume the ground plane is locally flat—an assumption that introduces lateral displacement errors of several metres on curved roads. Dubska et al. [11] achieve fully automatic calibration of roadside cameras from lane markings and vanishing points, but their model is restricted to a flat road plane; our pose solver extends this by additionally recovering camera height and pitch, and a second curved-centreline stage corrects residual lateral displacement on bends. To follow actual road geometry, our work introduces a road-centreline-following transform driven by a national road-network dataset [12], alongside a 4-point manual mode and a vanishing-point/lane auto-calibration path.

**ITS dashboards and data.** National ITS portals [13] publish CCTV streams and aggregated segment speeds, but typically as separate views; the aggregated speed is never fed back to correct the vision-derived estimate, leaving any geometric scale error unchecked. Our pipeline can *optionally* fuse the segment speed back as a soft cross-check, though by default it recovers the longitudinal scale geometrically (focal-free solve) and uses the ITS segment speed only as an independent comparison. Congestion grouping uses density-based clustering [14] and convex hulls [15]; service levels follow the Highway Capacity Manual [16]. The visualization uses deck.gl [17] over MapLibre [18], served by FastAPI [19].

#### A. Positioning Relative to State of the Art

We position this work against representative monocular traffic-analysis systems along the dimensions most relevant to uncalibrated public-infrastructure deployment. Table I summarises the comparison; the discussion below groups the prior art into three families and states explicitly where our system agrees with, differs from, and is limited relative to each.

**Closest setting: monocular CCTV speed estimation.** The most directly comparable line of work estimates vehicle speed from a single uncalibrated surveillance camera. Sochor et al. [20] established the de-facto benchmark for this task, BrnoCompSpeed, with LIDAR-derived ground-truth speeds, and report sub-km/h mean error on full-HD footage of predominantly straight motorway segments. Revaud and Humenberger [21] target the same problem under realistic

CCTV conditions—explicitly including low-resolution and non-straight roads—and, notably, evaluate on a dataset of Korean public CCTV streams with GPS-traced ground truth, the same infrastructure family this work builds on. Their calibration leverages the known 3D dimensions of vehicles; ours instead leverages the known 2D geometry of the road from the national node-link network. The two priors are complementary: a 3D-vehicle prior recovers metric scale per detection, whereas our centreline prior recovers road-following position over an extended corridor. We note that although the Revaud data *includes* non-straight roads, their method estimates speed without anchoring each vehicle to the road’s mapped curvature; it is robust *on* curved scenes but does not constrain positions *to* the curve, which is why we mark it partial ( $\Delta$ ) rather than full in the “Curved road” column of Table I. FARSEC [22] subsequently provided a reproducible pipeline evaluated on both BrnoCompSpeed and the Revaud CCTV set, underscoring that these two datasets are the accepted reference points for the task.

We emphasise an honest distinction. These methods report quantitative speed accuracy against instrumented ground truth, which our present evaluation does not: our speed figures (Table V) confirm plausible output rather than benchmark accuracy, and the per-camera `speed_scale` factors remain unconverged at the time of writing. We do, however, validate *localization* quantitatively against satellite-referenced landmarks (Section V), which is the dimension we claim as the contribution. Establishing comparable speed accuracy on the Revaud Korean-CCTV set or BrnoCompSpeed is the natural next step and is left as future work; the contribution claimed here is the map-anchored localization and self-calibrating pipeline, not a new speed-accuracy record.

**Automatic calibration from road geometry.** A second family recovers camera pose automatically from lane markings and vanishing points. Dubska et al. [11] achieve fully automatic calibration of roadside cameras from these cues, and Sochor et al. [20] refine scene scale by aligning 3D vehicle bounding boxes, reducing scale error substantially over earlier automatic methods. Both, however, model the road as a single flat plane. Our pose solver (`camera_pose.solve_pose`) follows the same vanishing-point/lane-edge strategy but additionally recovers camera height and pitch against a known road width, and—critically—a second centreline-following stage (Algorithm 1) corrects the residual lateral displacement that a flat-plane model leaves on bends. This is the dimension labelled “Curved road” in Table I, where prior monocular methods are marked unsupported.

**Dedicated sensors and on-vehicle GPS.** A third family sidesteps calibration entirely. Inductive-loop detectors, roadside radar, and on-vehicle GPS loggers measure speed and position directly, but require either physical roadside installation or instrumentation of the vehicles themselves. Our system adds no hardware to the road or to vehicles, re-purposing cameras that road authorities already operate; the trade-off is that all geometry is inferred and therefore bounded by the quality of the map and the perspective projection (Section VI).

TABLE I

COMPARISON WITH REPRESENTATIVE TRAFFIC-MONITORING APPROACHES. ✓ = FULLY SUPPORTED; × = NOT SUPPORTED; △ = PARTIALLY SUPPORTED; — = NOT APPLICABLE.

| Method                    | No road HW | No vehicle GPS | Auto-calib | Curved road | Speed est. | Real-time | Multi-cam | Open-source |
|---------------------------|------------|----------------|------------|-------------|------------|-----------|-----------|-------------|
| Loop / Radar sensors      | ×          | ✓              | —          | —           | ✓          | ✓         | △         | ×           |
| On-vehicle GPS            | ✓          | ×              | —          | ✓           | ✓          | ✓         | ✓         | △           |
| YOLO + flat homography    | ✓          | ✓              | ×          | ×           | △          | ✓         | ×         | ✓           |
| Dubská et al. [11]        | ✓          | ✓              | ×          | ×           | ✓          | ✓         | ×         | ×           |
| Sochor et al. [20]        | ✓          | ✓              | ✓          | ×           | ✓          | ✓         | ×         | ✓           |
| Revaud & Humenberger [21] | ✓          | ✓              | ✓          | △           | ✓          | ✓         | ×         | ✓           |
| <b>This work</b>          | ✓          | ✓              | ✓          | ✓           | ✓          | ✓         | ✓         | ✓           |

**Generic detection/tracking pipelines.** Finally, open-source YOLO+BoxMOT pipelines provide bounding boxes and track IDs but no geographic anchoring, calibrated speed, or support for uncalibrated cameras out of the box. Our system integrates these components and adds the localization, calibration, and self-correction layers that turn raw tracks into a map-anchored twin. Among the compared systems, ours is the only one supporting the full set of columns in Table I; we nonetheless note that this breadth reflects a systems-integration contribution rather than a single-metric advance, and that the per-axis novelty (curved localization, focal-free geometric scale recovery) is what we claim as new.

### III. SYSTEM DESIGN AND ARCHITECTURE

The backend (FastAPI/Uvicorn) and a React/deck.gl frontend communicate over REST and WebSocket. Two inference paths can produce per-frame analytics and *share a single detector*; a connection counter coordinates them so that the multi-object tracker is never driven by two interleaved frame streams (Fig. 1). A two-lock scheme inside the detector (`_state_lock` for tracker state, `_lock` for GPU inference) prevents data races between thread-pool threads, and module-level write locks in `main.py` and `roi_manager.py` atomize all JSON configuration read-modify-write operations.

Because running a separate detector per connection would multiply GPU memory use and violate the shared-tracker invariant, both paths feed a single shared detector; the connection counter ensures only one path drives the multi-object tracker at a time. The two paths are designed for complementary operating conditions.

**Browser path (/ws/detect).** When an operator opens a camera view, the frontend plays the HLS stream through a CORS proxy, captures canvas frames (JPEG,  $\leq 640$  px,  $\approx 33$  ms interval, at most two in flight), and sends them to the backend, which runs detection, tracking, localization, and analytics, then returns an annotated frame and broadcasts the analytics JSON to all clients.

**Server path (live\_loop).** Without an active browser client, the backend reads HLS directly with OpenCV/FFmpeg and runs the same pipeline. It is gated twice: it skips tracking while a browser detection client is active (to protect the shared tracker), and it skips all GPU work when no one is watching (the frontend reports tab visibility). Before opening any HLS URL the detector pre-resolves HTTP 302 redirects (`_`

`resolve_hls_url`), so FFmpeg receives a direct endpoint and does not silently fail on delivery-layer redirect chains.

Beyond the two inference paths, sustained operation requires housekeeping that neither path handles directly: HLS access tokens expire on a 30-minute cycle, the ITS speed reference drifts unless polled, and the congestion map must be periodically refreshed. **Background services.** Periodic loops therefore refresh expiring HLS tokens (30 min), poll the ITS segment speed for self-calibration (5 min), and snapshot all monitored cameras into a SQLite history while recomputing congestion clusters (30 s). To prevent the FOV polygon from flickering during auto-calibration, a slow EMA filter stabilizes near/far distance and road-width estimates, holding the initial accepted value fixed until sufficient samples accumulate. During the warm-up calibration phase the backend broadcasts a `calibrating` message (with elapsed time) every 30 frames so the frontend can show a progress badge, and emits an `auto_calibrated` message carrying the recovered focal length and the reprojection residual once the pose commits. ROI edits trigger an immediate GPS ring recomputation and broadcast to keep the map view synchronized. The resulting map view is shown in Fig. 2.

### IV. IMPLEMENTATION

#### A. Detection and Tracking

The detector loads the fastest available backend—TensorRT FP16 engine, then ONNX Runtime, then PyTorch—and exports `.pt`→`.engine` on first run when a GPU is present. A tracker *tier* is chosen by GPU memory: ByteTrack (CPU-only, no CUDA), OC-SORT (low VRAM  $\geq 3.5$  GB, occlusion-robust, no ReID), or BoT-SORT/DeepOC-SORT with OSNet appearance ReID on larger GPUs ( $\geq 6$  GB and  $\geq 10$  GB respectively). Without appearance ReID, two failure modes degrade practical tracking quality: detectors occasionally fire more than once on the same vehicle, inflating vehicle counts, and track identities reset after brief occlusions rather than recovering the original ID, causing discontinuous trails on the map. To suppress both effects, the non-ReID tiers include two lightweight robustness layers. `_dedup_tracks` merges duplicate boxes (IoU  $> 0.3$  or centre distance  $< 40$  px, keeping the older ID), while an `IDStabilizer` re-binds a vehicle’s previous ID after a brief miss using nearest-centre matching ( $\leq 80$  px) with a two-pass scheme and stale-mapping purge so display IDs stay stable and compact. Fig. 3 shows the detection and tracking overlay.

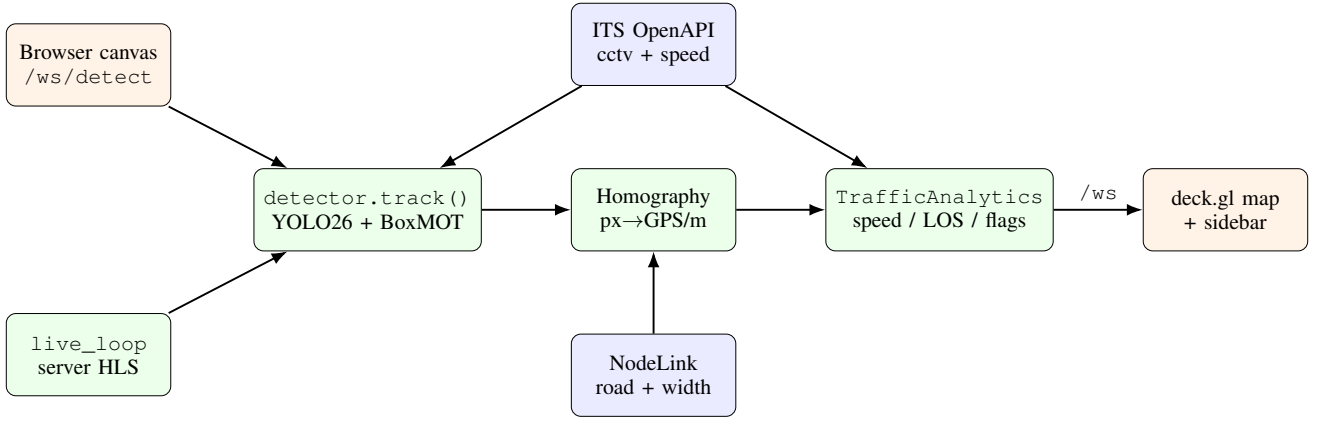


Fig. 1. System architecture. The browser path (`/ws/detect`) and the server path (`live_loop`) share one detector; a counter prevents concurrent tracker calls. External data (ITS CCTV/speed, NodeLink) feeds localization and self-calibration.

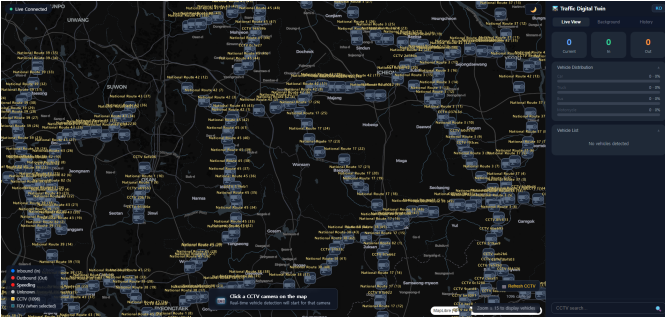


Fig. 2. Map view: public CCTV cameras as status-coloured icons on the vector map; the selected camera shows its field-of-view polygon and the located vehicles.



Fig. 3. Detection and tracking on a live CCTV frame: bounding boxes, vehicle class, and stabilized track IDs.

### B. Map-Anchored Localization

On camera switch we query the national node-link network: we snap the camera position perpendicularly onto the nearest road polyline, extract  $\pm 150$  m of centreline, and—because each direction is stored as a separate one-way link—we locate the reverse carriageway and average the two snaps (when 2–

40 m apart) to recover the true road centre. The chosen road’s speed limit, lane count, and width ( $\text{lanes} \times 2 \times \text{lane-width}$ ) are derived from NodeLink and attached to the camera.

A single flat homography maps pixels to a plane tangent to the camera’s snap point; on curved roads, this plane diverges from the true centreline and displaces vehicle markers laterally by several metres. To follow the actual road geometry, we instead introduce the *two-stage curved transform* (Algorithm 1). The metric homography  $H_m$  is derived from the same four image–GPS correspondences as the GPS homography, but maps pixel coordinates to local ENU metres rather than geographic coordinates, providing a metric workspace in which arc-length displacement can be computed accurately. Stage one maps the pixel to local metres via  $H_m$  and converts the result to a geographic coordinate via the snap-point anchor (ENUTOGPS); stage two projects that coordinate perpendicularly onto the centreline polyline (PERPPROJECT, Algorithm 2), obtaining the true arc length and a signed lateral offset directly from the nearest point on the curve, then re-applies the lateral offset perpendicular to the local road bearing at that arc. This replaces the earlier fixed-bearing decomposition ( $d_{\parallel} = dx \sin b + dy \cos b$ ), which under-measured the true arc length on bends and folded the road’s own curvature into the lateral term as a spurious offset; the perpendicular projection follows the bend and eliminates that artifact.

When no manual calibration exists, the system does *not* calibrate from a single frame. A single compressed CCTV frame rarely exposes the dashed lane markings needed to anchor scale—autocorrelation typically returns zero dashed-period observations—and calibrating from such a frame systematically overestimates speed. Instead, on camera switch the transformer enters a *warm-up* accumulation phase (`transform.start_warmup`): it samples a half-resolution grayscale road ROI every `CLEANPLATE_SAMPLE_S=0.5 s` into a bounded ring buffer (`CLEANPLATE_MAX_FRAMES=60`) and, every `WARMUP_EVAL_EVERY=30` frames, tests whether enough lane evidence has accumulated to commit early; a commit is forced after `WARMUP_MAX_S=90 s`. The *clean plate*—the per-pixel median

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**Algorithm 1** Two-stage curved pixel→GPS transform (perpendicular-projection)

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**Require:** pixel  $(u, v)$ ; metric homography  $H_m$ ; snap point  $p_m$  (ENU m); snap GPS  $(\phi_0, \lambda_0)$ ; centreline polyline  $C = \{(\phi_i, \lambda_i)\}$  with cumulative arc lengths  $\{\ell_i\}$

- 1:  $(x_m, y_m) \leftarrow H_m \cdot (u, v)$   $\triangleright$  pixel  $\rightarrow$  ENU metres
- 2:  $(dx, dy) \leftarrow (x_m, y_m) - p_m$   $\triangleright$  ENU offset from snap
- 3:  $(\phi_e, \lambda_e) \leftarrow \text{ENUToGPS}(\phi_0, \lambda_0, dx, dy)$
- 4:  $(arc, d_\perp) \leftarrow \text{PERPPROJECT}((\phi_e, \lambda_e), C)$   $\triangleright$  nearest point on centreline
- 5:  $(\phi_c, \lambda_c) \leftarrow \text{INTERPCENTERLINE}(arc)$
- 6:  $b_\ell \leftarrow \text{ROADBEARINGAT}(arc)$
- 7: **apply**  $d_\perp$  perpendicular to  $b_\ell$  at  $(\phi_c, \lambda_c)$
- 8: **return**  $(\phi, \lambda)$

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**Algorithm 2** PERPPROJECT: perpendicular projection onto centreline

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**Require:** query point  $q = (\phi, \lambda)$ ; centreline  $C = \{(\phi_i, \lambda_i)\}$  with arc lengths  $\{\ell_i\}$

- 1:  $d^* \leftarrow \infty$
- 2: **for** each segment  $(\phi_i, \lambda_i) \rightarrow (\phi_{i+1}, \lambda_{i+1})$  **do**
- 3:  $\vec{b} \leftarrow$  segment vector (local ENU metres)
- 4:  $\vec{p} \leftarrow q - (\phi_i, \lambda_i)$  (local ENU metres)
- 5:  $t \leftarrow \text{clamp}(\langle \vec{p}, \vec{b} \rangle / \|\vec{b}\|^2, 0, 1)$
- 6:  $s \leftarrow t \vec{b}$   $\triangleright$  foot of perpendicular
- 7:  $d \leftarrow \|\vec{p} - s\|$
- 8: **if**  $d < d^*$  **then**
- 9:  $d^* \leftarrow d$
- 10:  $arc \leftarrow \ell_i + t \|\vec{b}\|$
- 11:  $d_\perp \leftarrow \text{sign}(b_x p_y - b_y p_x) \cdot d$   $\triangleright$  signed lateral
- 12: **end if**
- 13: **end for**
- 14: **return**  $(arc, d_\perp)$

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of the buffered frames (`_build_clean_plate`)—removes transient vehicles and compression artefacts and reveals the static lane paint, so that the dashed-marking detector clears its gate ( $\geq \text{DASH\_MIN\_OBS}=3$  dashed-period observations). If a pose was saved for the camera in a previous session it is loaded immediately and the warm-up is skipped.

On commit (`commit_calibration`) the clean plate is upscaled to full resolution and a homography is estimated from its lane geometry: Canny edges in the lower image, Hough line filtering, multi-row sampling of left/right road edges, least-squares edge fits, and vanishing-point intersection. Direction (which way the camera looks along the road) is resolved by matching the *image* road-curve sign against the *map* curve sign, falling back to a vanishing-point heading test. A manual 4-point mode remains available for maximum accuracy. Fig. 4 shows an automatic calibration result.

The primary auto-calibration path is a road-model *pose solver* (`camera_pose.solve_pose`): the sampled lane edges, vanishing point, and NodeLink road width are fitted to a pinhole-camera model via `scipy.optimize.least_`

squares (soft-L1 loss). By default the focal length is fixed at  $\text{FOCAL\_RATIO} \cdot h = 1.2h$  and four parameters are solved (height  $H$ , pitch, yaw, lateral offset  $x_0$ ). **Focal-free solve.** When the clean plate yields enough dashed-marking observations ( $\geq \text{FOCAL\_FREE\_MIN\_OBS}=3$ , spanning at least  $\text{FOCAL\_FREE\_MIN\_ROW\_FRAC}$  of the frame height), the focal length is promoted to a fifth free parameter. The dashed-line *period* measured at several image rows anchors the *longitudinal* (depth) scale that road width alone cannot observe—recovering the focal length geometrically is the direct fix for the speed overestimation a fixed-focal assumption causes. This gate fires reliably on the median clean plate but rarely on a single live frame, which is precisely why the warm-up accumulator exists. The dashed-marking observations come from `lane_` `markings.detect_lane_` `markings`: after CLAHE contrast normalization it samples vertical strips at detected lane-line positions and takes the dominant 1-D autocorrelation period (accepted when the normalized peak  $\geq 0.15$ ), reporting one (row, period) pair per strip. A per-row reprojection residual gates the output ( $< \text{POSE\_RESIDUAL\_MAX\_PX}=8$  px); on success the solved pose is converted to four image-GPS corner pairs and used to build both homographies. The pose is then persisted per camera to `camera_pose.json` so each session refines the prior, with a cold-start fallback chain (saved prior  $\rightarrow$  vehicle-bbox rough pose  $\rightarrow$  GPS-grid). A complementary `vehicle_calib.json` accumulates a per-camera linear pixel-to-distance model ( $d = B \cdot y_{px} + C$ , implied vanishing-point  $y_v = -C/B$ ) fitted from observed vehicle bounding-box heights; it is restored on session start and updated online whenever sufficient samples accumulate. This provides a second, independent distance estimate—derived from the known apparent size of vehicles rather than from road geometry—that serves as a *self-consistency check* on the geometric homography. We use it as an auxiliary signal rather than a primary calibration: it does not replace the homography, but a large disagreement between the two flags an unreliable geometric fit. We do not claim this size-based scale as novel—recovering scene scale from known vehicle dimensions is established [20], [21]; here it functions only as a lightweight, training-free cross-check. The trapezoid heuristic above (pitch from the vanishing point, height from road width and a pinhole model) serves only as the fallback when the residual exceeds the quality gate or lane detection fails.

**Commit-once and lock.** On a successful commit the solved pose is saved, the vehicle-scale model is fitted once, and the transformer is *locked*: the pose is never re-solved and the scale refit is frozen (gated by the lock flag), so a calibrated camera cannot drift from later noisy frames. A `POST /recalibrate` request clears the lock, deletes the saved pose, and restarts the warm-up.

After GPS coordinates are assigned, a two-step post-processing pass reduces visual noise. Per-track EMA smoothing eliminates lateral jitter while preserving along-road motion; a jump threshold resets the filter after occlusion gaps to handle track re-appearances correctly. A lane-offset pass then laterally separates inbound and outbound vehicles perpendicular to the

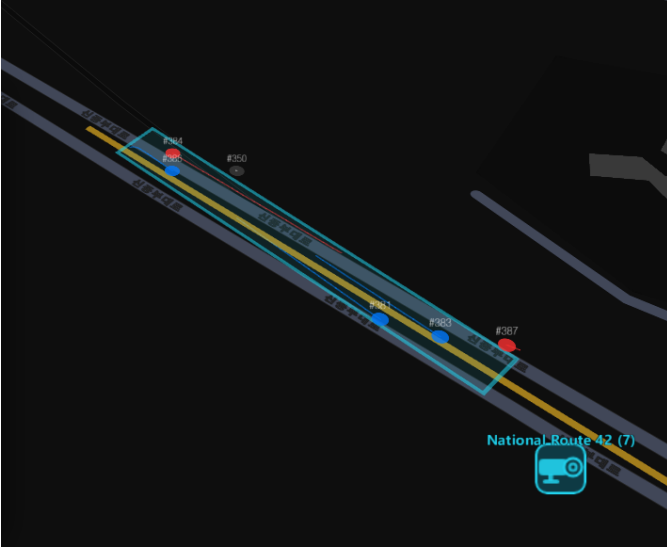


Fig. 4. Map view showing the estimated field-of-view footprint after automatic calibration: the cyan corridor follows the road centreline extracted from the NodeLink dataset.

local road bearing, placing each direction on its correct side of the road centreline for Korean right-hand traffic.

### C. Speed Estimation

Speed is distance over time, and both terms are error-prone in this setting. Homography-derived positions carry growing noise toward the far field, where small pixel displacements correspond to large ground-plane uncertainties. HLS streams compound the problem: segments are buffered and retransmitted with variable delivery latency, so consecutive frame-arrival times reflect network scheduling rather than true capture cadence, making wall-clock timing unreliable. Rather than treat these two error sources independently, we address them jointly: a monotonic clock derived from stream presentation timestamps (falling back to wall-clock) decouples timing from delivery jitter, while the five-stage estimator of Algorithm 3 filters the resulting distance-over-time signal using the constants in Table II.

All thresholds in Table II were empirically tuned on live ITS streams; sensitivity analysis is left as future work.

Three additional reliability mechanisms reduce systematic bias in the speed estimate. First, a depth-varying correction factor  $\kappa$  is computed from a per-camera vehicle apparent-size model (a linear fit of inverse apparent bbox width versus vertical pixel position), compensating for the depth-dependent projection error of the homography without introducing frame-to-frame position discontinuities. Second, vehicles whose GPS position lies more than 100m from the camera snap point (`SPEED_TRUST_MAX_DEPTH_M`) are flagged `speed_reliable=False`: their speed is still displayed, but excluded from the frame average, the ITS calibration input, and speeding alerts, preventing far-field pixel noise from biasing aggregate statistics. Third, when the frame average is formed over the reliable vehicles, a robust cross-vehicle filter (`_`

### Algorithm 3 Per-track speed estimation (analytics.\_speed)

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**Require:** window  $W$ , prior EMA  $v_e$ ,  $\sigma \leftarrow \text{speed\_scale}$ ,  $\kappa \leftarrow \text{depth\_corr}(y_{\text{bbox}})$

- 1: **if** duplicate position **then** skip ▷ dedup
- 2: **if**  $\text{step} > (V_{\text{max}}/3.6\sigma) dt \cdot 3.0$  **then** clear  $W$  ▷ teleport reset
- 3: **else if**  $\text{step} > (V_{\text{max}}/3.6\sigma) dt \cdot 1.5$  **then** drop sample, keep  $W$  ▷ jump guard
- 4: **if**  $dt > 2\text{ s}$  **then** clear  $W$  ▷ track gap
- 5: fit  $\hat{v}$  by OLS on  $W$  (0.7 s); **if**  $\text{disp}(W) < 0.5\text{ m}$  **then**  $\hat{v} \leftarrow 0$  ▷ OLS
- 6:  $v_e \leftarrow (1-\alpha)v_e + \alpha\hat{v}\kappa\sigma$ ,  $\alpha=0.35$ ; reject spike; decay on stop ▷ EMA
- 7: **output**  $v_e$ ;  $\text{is\_speeding} \leftarrow \text{reliable} \wedge v_e > 1.10 \cdot \text{limit}$  ▷ output

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TABLE II  
KEY SPEED/ANALYTICS CONSTANTS (`CONFIG.PY`).

| Parameter                      | Value                                              |
|--------------------------------|----------------------------------------------------|
| Speed window                   | 0.7 s ( $W=\text{round}(0.7s \times \text{FPS})$ ) |
| EMA $\alpha$                   | 0.35                                               |
| Jitter threshold               | 0.5 m                                              |
| Min speed (else 0)             | 5 km/h                                             |
| Max plausible $V_{\text{max}}$ | 180 km/h                                           |
| GC grace                       | 30 frames                                          |
| Bottleneck dwell               | 150 frames ( $\approx 5\text{ s}$ )                |
| Parked dwell                   | 300 frames ( $\approx 10\text{ s}$ )               |
| Speeding threshold             | $v_e > 1.10 \times \text{limit}$                   |
| LOS thresholds A–E             | $\leq 3, 6, 9, 12, 15$ (else F)                    |
| Background busy                | $> 6$ vehicles                                     |
| Background congested           | $> 14$ vehicles                                    |
| Cluster $\varepsilon$          | 500 m (haversine)                                  |

`reject_speed_outliers`) drops any sample more than  $K=\text{SPEED\_OUTLIER\_MAD\_K}=3$  scaled median-absolute-deviations (1.4826 MAD) from the median (applied only with at least three vehicles), so a single ID-swap or homography spike cannot pull the average that also feeds the `speed_scale` statistics.

### D. Optional ITS Scale Cross-Reference

The focal-free pose solve (Section IV-B) recovers the longitudinal scale geometrically from the dashed-marking period, so it is the primary mechanism for absolute speed scale; a residual error can remain only on cameras where dashed markings are not observable. As an *optional* refinement—**disabled by default** (`ITS_DRIVES_SCALE=false`)—the system can maintain a per-camera `speed_scale` factor that multiplies the geometric speed estimate (Algorithm 3, line 6), updated online using the ITS official segment speed as a soft reference. With the option off (the default and evaluated configuration), the ITS segment speed is shown only as an independent comparison (Section IV-F) and is never fed back into the estimate. When enabled, the ITS signal carries inherent uncertainty—aggregated over  $\approx 1\text{ km}$  segments at 5-minute



intervals, the reported direction may not match the camera’s field of view—so the loop adapts slowly:

$$target = scale \cdot \frac{v_{ITS}}{v_{ours}}, \quad scale \leftarrow (1 - \alpha_s) scale + \alpha_s target, \quad (1)$$

with an adaptive  $\alpha_s$  schedule and a per-update  $\pm 10\%$  step clamp in addition to the absolute  $[0.3, 5.0]$  bounds. For cameras without a previously saved factor (first session),  $\alpha_s$  decreases from 0.15 (updates 1–2) through 0.05 (updates 3–4) to 0.01 thereafter, enabling fast initial convergence while damping later oscillation. For cameras whose factor was restored from `speed_scale.json`,  $\alpha_s$  is fixed at 0.01 to preserve the accumulated calibration. The calibration uses a 10-minute rolling window of continuous, per-frame speed observations, requiring at least 50 frames with valid vehicle detections to trigger the correction, and a coefficient-of-variation guard ( $\leq 0.4$ ) to skip volatile traffic. The factor persists per camera and accumulates across sessions without resetting the geometric calibration (Section VI).

#### E. Aggregation, Congestion, and History

Assigning inbound versus outbound direction from a single frame’s pixel displacement is unreliable when a vehicle moves slowly, stops briefly, or travels nearly perpendicular to the camera view—the displacement vector is then too short or misaligned to determine sign robustly. To accumulate directional evidence across frames, each vehicle’s inter-frame displacement is projected onto the road bearing axis and smoothed with an EMA deadzone filter that commits a direction only when the smoothed component exceeds an ambiguity threshold. When road bearing is unavailable, a horizontal LineZone crossing serves as a fallback and also provides In/Out counts. The road bearing itself is auto-refined online (`refine_bearing`): per-frame vehicle-flow vectors are accumulated with double-angle statistics (free of the  $180^\circ$  direction ambiguity) and, after enough samples, blended into the working bearing by a slow EMA. Dwell counting flags bottlenecks and parked vehicles (the latter remembered by pixel position so they survive ID changes), and a Level-of-Service grade is assigned from the active vehicle count. Background monitoring polls additional cameras every 8 s with detection only; cameras with more than 6 vehicles are marked *busy* and those with more than 14 are marked *congested*; such cameras are then clustered with a haversine density-based scan ( $\epsilon=500$  m) and outlined with a convex hull. A WAL-mode SQLite store records 30 s snapshots (14-day retention) and serves bucketed time series, peak-time detection, and CSV export. Fig. 5 shows the congestion overlay.

#### F. Visualization

The `deck.gl` map layers congestion polygons, vehicle trails, the FOV polygon, status-coloured camera icons, and vehicle markers. The FOV polygon uses one of three strategies in priority order: (1) the manual calibration ring (`calibGpsRing`), the exact four GPS corners entered by the user; (2) a curved polygon computed by walking the road centreline and laterally offsetting by half the road width (`computeRoadCorridorPolygon`), so the footprint

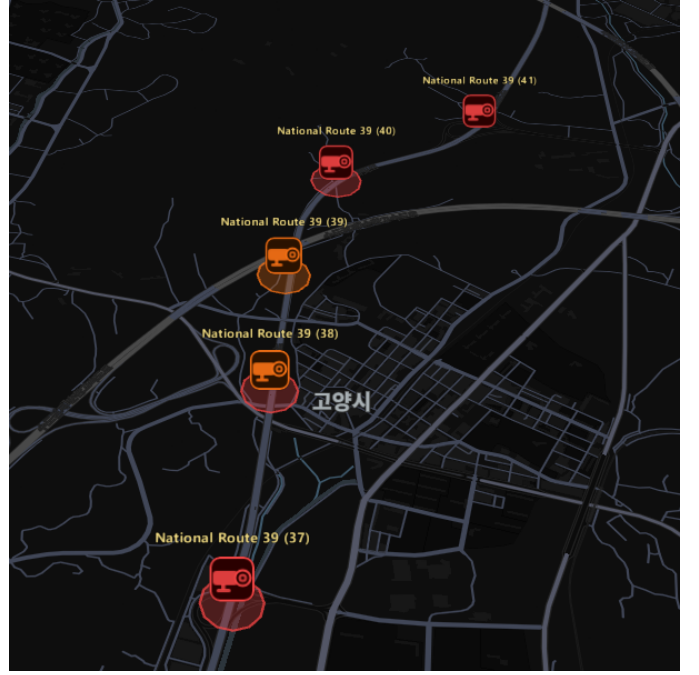


Fig. 5. Camera-level congestion: monitored cameras are recoloured (normal/busy/congested) and adjacent congested cameras are grouped into a severity-shaded region.

follows real road curvature rather than a flat projection; or (3) a straight-sided rectangle derived from near/far distance and road width (`computeCalibPolygon`), falling back to a default  $70^\circ$  trapezoid (`computeFovPolygon`) when uncalibrated.

The sidebar’s vehicle list provides *direction-filter tabs* (All / Inbound / Outbound) that filter the displayed rows by the movement-vector-based direction; the per-direction vehicle count is shown in each tab badge, and the min/avg/max speed summary is recomputed from the filtered set. Two collapsible cards (Auto Calibration Estimate, ITS Speed Comparison) include an info button that opens a modal explanation of each metric, adapting to the selected map theme. During warm-up the Auto Calibration card shows a progress badge (*Calibrating (Ns)*) driven by the `calibrating` broadcast; after the pose commits it displays the recovered focal length and a colour-coded reprojection residual, and exposes a *Re-calibrate* button that issues `POST /recalibrate` to clear the lock and restart the warm-up.

## V. EVALUATION

Measurements are produced from the real pipeline in two ways. While the system runs (`make dev`), an in-process collector records per-frame stage latency, tracking, speed, and detection statistics and flushes `logs/eval_*.csv` and `logs/eval_summary.json` every 30 s (also on demand via `GET /eval/report`); an offline harness (`backend/evaluate.py`) reproduces the same metrics over a live HLS source for controlled benchmarks. The system auto-selects the inference backend (TensorRT  $\rightarrow$  ONNX Run-

TABLE III  
INFERENCE BACKEND COMPARISON (OC-SORT TRACKER FIXED,  $\approx 60$  S PER BACKEND, RTX 4070 LAPTOP, NATIONAL ROUTE 3 SEONGNAM DAEWON IC, DENSE DAYTIME TRAFFIC). TRACK STAGE INCLUDES YOLO + BOXMOT END-TO-END.

| Backend        | Mean (ms) | p95 (ms) | FPS  |
|----------------|-----------|----------|------|
| TensorRT FP16  | 16.3      | 19.9     | 59.6 |
| PyTorch (CUDA) | 28.0      | 33.1     | 35.1 |
| ONNX Runtime   | 29.3      | 38.1     | 33.4 |

TABLE IV  
TRACKER COMPARISON (TENSORRT FP16 BACKEND FIXED,  $\approx 60$  S EACH, NATIONAL ROUTE 3 SEONGNAM DAEWON IC, DENSE DAYTIME TRAFFIC). BoT-SORT OVERHEAD IS FROM OSNET APPEARANCE REID; BYTETRACK FRAGMENTS HEAVILY WITHOUT RE-IDENTIFICATION.

| Tracker   | Mean (ms) | p95 (ms) | FPS  | IDs | ID ev. | Life (fr.) |
|-----------|-----------|----------|------|-----|--------|------------|
| ByteTrack | 43.9      | 61.2     | 22.4 | 946 | 2940   | 13.1       |
| OC-SORT   | 16.3      | 19.9     | 59.6 | 272 | 1609   | 121.1      |
| BoT-SORT  | 60.7      | 78.6     | 16.3 | 79  | 214    | 176.0      |

IDs = unique track IDs; ID ev. = ID creation events; Life = mean lifetime.

time  $\rightarrow$  PyTorch) and tracker tier (ByteTrack / OC-SORT / BoT-SORT / DeepOC-SORT by GPU VRAM).

**Hardware.** All timing figures are from a single machine: NVIDIA GeForce RTX 4070 Laptop GPU (8 GB VRAM, compute capability 8.9), driver 610.47, TensorRT 10.16, ONNX Runtime 1.25, Windows 11. Absolute latency and FPS are hardware-specific; the relative ordering across tiers and backends is the transferable result.

**Run conditions.** Model YOLO2.6m, source: National Route 3 Seongnam Daewon IC (live ITS HLS, dense daytime traffic), measured for  $\approx 60$  s per configuration (20-frame warmup excluded), so the frame count per run varies with throughput. Backend and tracker are varied across runs as shown in Tables III and IV; all other conditions are held constant. Each table is an independent measurement run over a different 60 s window, so the shared TensorRT + OC-SORT operating point differs slightly across Tables III–IV, and the tracking-stability counts vary with the traffic present at each capture time. The latency/tracker runs use the uncalibrated grid homography (timing is independent of calibration quality); the tracking-stability, speed, and detection figures in Table V are from a separate, longer *pose-calibrated* live run on the same camera (6,877 frames), included to confirm the pipeline produces plausible output, not as ground-truth benchmarks.

TensorRT FP16 achieves the lowest track-stage latency (16.3 ms mean),  $1.8\times$  faster than ONNX Runtime (29.3 ms) and  $1.7\times$  faster than PyTorch (28.0 ms) on the same GPU. The harness additionally breaks end-to-end time into pipeline stages; the homography transform and analytics stages each contribute under 0.5 ms (median 0.06 ms and 0.43 ms respectively), confirming that the localization and speed-estimation layers add negligible overhead to the dominant tracking stage. With the backend fixed to TensorRT, the dense daytime scene separates the trackers sharply. OC-SORT is simultaneously the fastest (16.3 ms) and far more stable than ByteTrack, whose lack of

TABLE V  
TRACKING STABILITY, SPEED DISTRIBUTION, AND DETECTION BREAKDOWN (TENSORRT + OC-SORT RUN, 6,877 FRAMES, NATIONAL ROUTE 3 SEONGNAM DAEWON IC).

| Metric                                               | Value                                    |
|------------------------------------------------------|------------------------------------------|
| Frames processed                                     | 6,877                                    |
| Unique tracks                                        | 556                                      |
| Track creation events (re-entries + re-acquisitions) | 2,164 ( $3.89\times$ unique tracks)      |
| Mean track lifetime (frames)                         | 96.7                                     |
| Median track lifetime (frames)                       | 63.5                                     |
| Max track lifetime (frames)                          | 611                                      |
| Vehicle speed – mean (km/h)                          | 85.7                                     |
| Vehicle speed – median (km/h)                        | 79.9                                     |
| Vehicle speed – p95 (km/h)                           | 175.3 <sup>†</sup>                       |
| % moving observations                                | 84.4 %                                   |
| Total speed observations                             | 45,393 of 53,772 detections <sup>‡</sup> |
| Detections – car                                     | 42,866                                   |
| Detections – truck                                   | 10,088                                   |
| Detections – bus                                     | 818                                      |

<sup>†</sup>Far-field outliers; median 79.9 km/h is representative.

<sup>‡</sup>Speed requires  $\geq W$  track-history frames; single-frame detections excluded.

re-identification fragments the heavy traffic into 946 short-lived tracks (2,940 ID-creation events, 13-frame mean lifetime) and inflates association cost to 43.9 ms. BoT-SORT’s OSNet appearance ReID yields the most stable tracks (79 IDs, 214 ID events, 176-frame mean lifetime) but is the slowest (60.7 ms mean, 78.6 ms p95) due to the embedding network. OC-SORT therefore gives the best stability/latency trade-off for this workload, while BoT-SORT is preferable when maximum ID persistence matters and the GPU budget allows.

**Optional ITS scale cross-reference (archived).** The default pipeline recovers the speed scale geometrically (the focal-free solve), so the ITS `speed_scale` loop is an optional, off-by-default refinement rather than a core mechanism. For completeness we report an archived snapshot from earlier operation with the loop enabled: Table VI summarises the factors accumulated across the cameras monitored in that mode. The distribution is informative as a diagnostic—it shows how far the raw geometric projection deviated per camera before any focal-free correction: the bulk cluster within  $\pm 20\%$  of 1.0 (geometric calibration roughly correct), while the tails reflect steep-pitch / far-field cameras that compress apparent displacement (below 0.8) and wide-angle / low-mounted cameras that exaggerate it (above 1.2). It is *not* part of the default evaluated configuration.

**Satellite-referenced localization accuracy.** To measure localization directly—without instrumented vehicles—we adopt a landmark-based protocol in the spirit of Revaud and Humenberger’s `evaluate_homographies` [21], which scores the homography against fixed ground-truth references and so isolates geometry from detector and tracker noise. Rather than GPS-tracked cars, we use fixed road landmarks (lane-dash ends, painted markings, structural points) whose true coordinates are read from high-resolution aerial imagery; for each landmark



TABLE VI

DISTRIBUTION OF `SPEED_SCALE` FACTORS FROM AN *archived* RUN WITH THE OPTIONAL ITS LOOP ENABLED (OFF BY DEFAULT IN THE CURRENT PIPELINE, WHICH RECOVERS SCALE GEOMETRICALLY VIA THE FOCAL-FREE SOLVE). VALUES BELOW 1.0 INDICATE GEOMETRIC UNDERESTIMATION OF DISPLACEMENT (CAMERA AXIS OBLIQUE TO THE ROAD); ABOVE 1.0, OVERESTIMATION. SHOWN AS A PER-CAMERA PROJECTION-BIAS DIAGNOSTIC, NOT A DEFAULT-MODE RESULT.

| Range                      | Cameras | Share  |
|----------------------------|---------|--------|
| < 0.5                      | 4       | 10.8 % |
| 0.5–0.8                    | 10      | 27.0 % |
| 0.8–1.2                    | 19      | 51.4 % |
| > 1.2                      | 4       | 10.8 % |
| Min / Max: 0.34 / 2.25     |         |        |
| Mean / Median: 0.90 / 0.85 |         |        |

we query the running backend in its current (pose-calibrated) state and compare its `pixel_to_gps` output against ground truth, decomposing the error into along-road (longitudinal) and lateral (off-road) components—the two modes the paper separates. The landmarks are independent of any calibration input, so the figures reflect generalization, not fit.

Table VII reports two pose-calibrated cameras on national routes, each scored under the flat homography and the curved-centreline transform. Two facts stand out. First, the system places vehicles with low *lateral* error—1.8 m on the better-conditioned camera (Rte. 6, 5 px reprojection residual) and 10.3 m on the more obliquely-mounted one (Rte. 3, 6.4 px)—confirming that the map-anchored geometry puts vehicles on the correct part of the carriageway. Second, the lateral and longitudinal components behave exactly as the orthogonality analysis in Section VI predicts: the longitudinal error is far larger (29–40 m), set by the residual focal-length scale uncertainty, while the lateral error is small and is the only component the curved transform can act on.

On these well-calibrated cameras the curved transform is *neutral*: it changes lateral error by under 0.2 m, because when the metric homography is accurate the flat plane already recovers ground positions correctly on a planar curve (every point of the curve lies on the fitted  $z=0$  plane). The curved transform is therefore best understood not as a uniform accuracy improvement but as a lateral-conformance safeguard that constrains markers and the FOV polygon to the road’s mapped shape without degrading an already-accurate flat fit—and it engages meaningfully only when that flat fit is weak. The dominant residual on both cameras is longitudinal, which no geometric transform corrects; it is the target of the `speed_scale` loop (Section VI).

## VI. DISCUSSION AND LIMITATIONS

The homography is accurate inside the calibration quadrilateral and extrapolates with growing error toward the far field; we mitigate this with the dual-matrix calibration, the regression window, and lane-based auto-calibration, but a residual error remains on poorly observed cameras. Automatic calibration recovers the focal length geometrically (the focal-free pose

TABLE VII

SATELLITE-REFERENCED LOCALIZATION ERROR (METRES) ON TWO POSE-CALIBRATED CAMERAS, FLAT HOMOGRAPHY VS. CURVED-CENTRELINE TRANSFORM, DECOMPOSED INTO LATERAL (OFF-ROAD) AND LONGITUDINAL (ALONG-ROAD) COMPONENTS. LANDMARKS ARE INDEPENDENT OF CALIBRATION POINTS; GROUND TRUTH FROM AERIAL IMAGERY, QUERIED AGAINST THE LIVE BACKEND. THE ERROR IS DOMINATED BY THE LONGITUDINAL COMPONENT (FOCAL-SCALE UNCERTAINTY, SECTION VI); THE CURVED TRANSFORM ACTS ONLY ON THE SMALL LATERAL COMPONENT AND IS NEUTRAL HERE BECAUSE THE FLAT FIT IS ALREADY ACCURATE. LANDMARK COORDINATES AND THE QUERY SCRIPT ARE TO BE DEPOSITED IN THE PUBLIC REPOSITORY TO SUPPORT REPRODUCIBILITY.

| Camera (calib. residual) | Method | Lateral | Along | Total |
|--------------------------|--------|---------|-------|-------|
| Rte. 3 (6.4 px)          | flat   | 10.3    | 40.3  | 41.6  |
|                          | curved | 10.1    | 40.4  | 42.4  |
| Rte. 6 (5 px)            | flat   | 1.8     | 29.2  | 29.3  |
|                          | curved | 1.8     | 29.1  | 29.2  |

Lateral / Along = MAE of off-road / along-road error; 4–5 landmarks per camera.

solve, Section IV) only when dashed lane markings are visible on the warm-up clean plate; otherwise it falls back to a fixed focal length of  $1.2 \times$  image height (unknown intrinsics), leaving a  $\pm 15$ – $20$  % FoV-dependent scale uncertainty, and to an approximate grid when lane detection fails entirely (night, rain, dense traffic). Road width is inferred from lane count and assumed bidirectional, so one-way roads are over-wide. The nearest node-link road is occasionally the wrong one (e.g., a camera near both a national route and an expressway), which displaces the FOV polygon and vehicle GPS; the CCTV-name road-hint reduces but does not eliminate this. Finally, congestion is clustered at the camera level because background cameras expose only counts, not per-vehicle speeds.

**Error structure of curve localization, and the roles of the two corrections.** The two-stage centreline transform (Algorithm 1) and the per-camera `speed_scale` loop (Section IV-D) are easily conflated as a single “accuracy” mechanism. A controlled synthetic analysis—projecting vehicles onto a circular-arc centreline of known curvature with exact ground-truth positions and recovering them through both the flat homography and Algorithm 1—yields a more precise account of what each component does.<sup>1</sup>

Two findings emerge. First, when the metric homography  $H_m$  is geometrically accurate, the flat homography already recovers ground positions correctly on a planar curve, since every point of the curve lies on the same  $z=0$  plane the homography was fitted to; in this regime the centreline transform changes accuracy negligibly. The displacement that matters in practice comes instead from the focal-length assumption in the fixed-focal calibration regime ( $f_y = 1.2h$ , a  $\pm 15$ – $20$  % FoV-dependent scale error)—which the focal-free solve removes

<sup>1</sup>Synthetic conditions: pinhole camera at 10 m height,  $18^\circ$  downward pitch, focal length fixed at  $f = 864$  px ( $1.2 \times$  a 720 px frame height, modelling the fixed-focal fallback regime), snap point 10 m ahead of the camera; circular-arc roads of radius  $R \in \{80, 120, 200, 300, 500, 1000, \infty\}$  m over a 160 m corridor; vehicles placed with up to  $\pm 3.5$  m lane offset and 1.5 px detection noise. Errors are exact since ground-truth positions are known by construction.

when dashed markings are observable on the clean plate—and it falls almost entirely in the *longitudinal* (depth) direction rather than laterally. Second, the original fixed-bearing decomposition in Algorithm 1 constituted a coordinate artifact: resolving the ENU displacement against the snap point’s single fixed bearing under-measured the true arc length on a bend and folded the road’s own curvature into the lateral term as a spurious offset, displacing markers by tens of metres at small radii in simulation. Replacing that decomposition with the perpendicular projection (PERPROJECT, Algorithm 2)—which follows the bend and recovers the true arc length—removes the artifact and reduces the simulated off-road lateral error to below one metre across the tested curvature range, while leaving the straight-road case unchanged.

These results show that the lateral and longitudinal error modes are geometrically orthogonal and require different corrections. The satellite-referenced measurements in Section V (Table VII) confirm this structure on real cameras: on both pose-calibrated cameras the error is dominated by the longitudinal component (29–40 m, set by focal-scale uncertainty), while the lateral component is small (1.8–10.3 m) and is the only axis the curved transform alters—changing it by under 0.2 m when the flat fit is already accurate. The real data thus supports the orthogonality claim and the role assignment (centreline geometry for lateral placement; the focal-free solve and the `speed_scale` loop for longitudinal scale), while showing that the curved transform is a safeguard that does not degrade a good flat fit rather than a uniform accuracy gain. The centreline transform addresses only the *lateral* component: it constrains vehicle markers and the FOV polygon to the road’s true shape on bends, but it cannot recover absolute longitudinal scale, since the along-road distance inherits whatever scale error the homography carries. Longitudinal scale is recovered *geometrically* by the focal-free pose solve, which reads the focal length from the dashed-marking period when those markings are observable on the clean plate. For cameras where they are not, the residual scale error can *optionally* be absorbed by the ITS-referenced `speed_scale` loop (disabled by default), which nudges a per-camera multiplicative factor toward the official segment speed. The mechanisms are therefore complementary rather than redundant: the centreline geometry fixes *where on the road* a vehicle is placed, and the focal-free solve fixes *how far* the geometry believes it has travelled, with the optional ITS loop available only as a fallback cross-check. This separation also disciplines evaluation: lateral road-alignment can be judged geometrically (markers either lie on the carriageway or they do not), whereas longitudinal accuracy is only meaningful against a speed or distance reference.

When enabled, the optional ITS loop uses an adaptive learning rate (0.15 on the first two updates, decaying to 0.01 thereafter, polled every 5 min) and treats the ITS segment speed as an imprecise soft reference, since ITS figures reflect 1-km road-segment averages that may not align with the camera’s exact observation zone or direction. It is *off by default*: the focal-free geometric solve is the primary scale mechanism, so the pipeline does not depend on ITS convergence. When

the loop is used, the accumulated factors should be read as a per-camera projection-bias diagnostic rather than a definitive calibration.

Because the system derives all geographic positions from map geometry rather than on-vehicle GPS receivers, localization accuracy is bounded by the quality of the NodeLink road database and the precision of the perspective projection; errors in map alignment or road-shape approximation propagate directly into vehicle GPS coordinates. Additionally, a camera’s effective field of view is strongly shaped by its mounting angle: a steeply pitched installation sees a narrow, elongated strip of road, producing a thin FOV polygon on the map and limiting the usable observation window even after geometric calibration.

We also state plainly the remaining evaluation gap. We now report quantitative *localization* accuracy against an external reference—the satellite-referenced landmark measurements of Section V—but this validation is still limited in scope: two cameras with a handful of landmarks each, bounded below by the 1–3 m precision of the aerial imagery itself, and not yet a large-scale study. We do *not* yet report quantitative *speed* accuracy against an external reference; the speed figures in Table V confirm plausible output rather than benchmark accuracy. The landmark protocol naturally extends to more cameras and denser landmark sets, and speed can be evaluated on the public Korean-CCTV and BrnoCompSpeed references [20], [21]. We leave this broader quantitative validation—more cameras, and a speed benchmark—as the immediate next step; the contribution claimed here is the map-anchored localization and self-calibrating pipeline, not a speed-accuracy result.

A structural limitation is the system’s dependence on national ITS infrastructure: camera streams and segment speeds are fetched from a government-operated open API [13] whose availability and authentication policy are outside the operator’s control and may be restricted or discontinued. Camera placement and viewing angle are fixed by road authorities and cannot be adjusted; some installations are positioned too far or too close to capture a useful road segment, and extreme viewpoints cannot be recovered through calibration. Adverse conditions—low illumination, rain, lens contamination, and hardware faults—can suppress the visual signal entirely, leaving the pipeline with no basis for detection or calibration. These constraints are inherent to re-using existing infrastructure without modification.

## VII. CONCLUSION

We built a real-time traffic digital twin that localizes vehicles on a map from uncalibrated public CCTV and estimates their speed without ground truth. The key ideas are a road-centreline-following two-stage transform, a robust five-stage speed estimator, and geometric speed-scale recovery via a focal-free pose solve (with an optional, off-by-default ITS segment-speed cross-check), wrapped in a resilient dual-path pipeline with congestion analytics and a historical store. The automatic calibration pipeline—a warm-up clean-plate pose solver that recovers focal length and scale geometrically—substantially reduces the per-camera manual setup required to

deploy across many cameras, and the system needs no external speed reference to operate.

A distinguishing property is that the system requires *no GPS instrumentation on vehicles*: conventional traffic monitoring relies on embedded GPS loggers or dedicated roadside sensors, both of which demand significant infrastructure investment. By re-purposing cameras that road authorities already operate, the system tracks multiple vehicle classes simultaneously across multiple views with no additional hardware cost. As public CCTV coverage continues to expand and object detectors advance, this class of vision-only traffic twin becomes increasingly practical at scale. With wider network integration, the same localization principle—positioning objects through perspective projection rather than satellite signal—could extend to GPS-denied environments such as tunnels, underground parking, and indoor logistics facilities; the per-camera track-ID scheme further enables cross-camera vehicle re-identification for a range of fleet management and security applications.

The source code is publicly available at <https://github.com/MinsooAhn02/DigitalTwin>.

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